Test Strategy Document – Web UI Role-Based Access Control

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1. Introduction

The Web UI under test provides two distinct user categories (roles) that dictate what configuration functions a logged-in user may invoke:

Role	Permitted Functions (examples)
Admin	• Manage user-related modules (Parent-Control, Firewall, Access-Control Lists)
Engineer	• Update firmware, change ISP configuration, view system logs, perform diagnostics

The purpose of this strategy is to define **how** we will verify that the UI enforces these role-based permissions correctly, and to outline the measurement criteria (pass/fail) that will be used to assess compliance.

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2. Scope

In-Scope	Out-Of-Scope
• Role-based navigation menus and page access	• Backend API unit-tests (covered by separate integration test suite)
• CRUD operations on Admin-only modules (Parent-Control, Firewall)	• Mobile native app UI (only web UI is covered)
• Engineer-only actions (Firmware upload, ISP config)	• Non-functional performance tests (covered in performance test plan)

• Session handling, logout, and timeout behavior	• Third-party integrations not exposed through the UI
• Negative tests (attempted access to unauthorized pages)	• Internationalisation / localisation testing

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3. Test Levels & Types

Test Level	Test Type	Objective	Primary Artefacts
Component	Functional UI tests (Selenium/WebDriver)	Verify each UI component enforces role permissions	Test Cases, Automation Scripts
System	End-to-End workflow tests (Cypress)	Validate whole user journey from login to configuration commit	Test Suites, Test Data
Acceptance	Role-Based Access Acceptance Tests (RB-AAT)	Confirm the system meets the business requirements for Admin & Engineer	Acceptance Test Report
Security	Access Control Penetration Tests (OWASP ZAP)	Ensure no privilege escalation via URL manipulation, API calls, or DOM tampering	Security Test Log

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4. Test Design Approach

1. **Requirement Mapping** – Each functional requirement (e.g., "Admin can enable Parent-Control") is traced to a unique test case ID.
2. **Equivalence Partitioning** – Separate partitions for authorized vs. unauthorized actions per role.
3. **Boundary Analysis** – For inputs that involve numeric ranges (e.g., firmware version numbers).
4. **Negative Path Testing** – Direct URL access, hidden UI elements, and API calls attempted without proper session tokens.
5. **Data-Driven Testing** – Test data (user credentials, config payloads) stored in external CSV/JSON files to drive the same test script across multiple roles.

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5. Test Environment

Component	Version	Configuration
Web Browser	Chrome 124, Firefox 126, Edge 124	Latest stable, cleared caches before each run
Test Harness	Selenium 4.15 + TestNG 7.7	Grid nodes: 3 (Windows, Linux, macOS)
Application Server	Tomcat 10.1	Staging URL: `https://staging.example.com/ui`
Database	PostgreSQL 15	Seeded with role-specific test users (see §6)
Network	Isolated VLAN	No external internet access (to prevent firmware download leakage)

All environments are version-controlled via IaC (Terraform + Ansible).

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6. Test Data

Data Set	Description	Example Values
Admin Users	Valid admin credentials, belonging to “admin” group	`admin1 / P@ssw0rd!`
Engineer Users	Valid engineer credentials, belonging to “engineer” group	`eng1 / Eng!2026`
Invalid Users	Credentials without any role or with corrupted token	`invalid / wrong`
Firmware Package	Small dummy firmware image (1MB) for upload test	`firmware_v1.2.bin`

ISP Config Payload	JSON payload representing ISP settings	`{ "dns": "8.8.8.8", "gateway": "192.168.0.1" }`
Parent-Control Rules	Simple rule set for enabling/disabling	`{ "block": ["social.com"], "schedule": "night" }`

All data files are stored in the

`test-data/`

repository folder and are version-controlled.

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7. Test Cases

Below is a ***representative*** *matrix. Full test case catalog (≈ 150 cases) is attached in*

`TestCases_RBAC.xlsx`

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TC-ID	Role	Feature	Test Step Summary	Expe
TC-ADMIN-001	Admin	Login	1. Navigate to login page 2. Enter admin credentials 3. Click **Login**	Dashbo **Admin** displa
TC-ADMIN-010	Admin	Parent-Control (Create)	1. From **Settings → Parent-Control** click **Add Rule** 2. Fill rule fields 3. Click **Save**	Rule a list, `200 OK`
TC-ADMIN-020	Admin	Firewall (Edit)	1. Open **Firewall → Rules** 2. Edit existing rule 3. Save	Modif persis error
TC-ADMIN-030	Admin	Engineer-Only	1. Navigate directly to	Access

		Feature – Firmware Upload	**Engineer → Firmware** URL 2. Observe page	or re dashb *Insuf permi* messa
TC-ENG-001	Engineer	Login	Same as TC-ADMIN-001 but with Engineer credentials	Dashb **Engin** displa
TC-ENG-010	Engineer	Firmware Upload	1. Go to **Engineer → Firmware** 2. Click **Upload** 3. Select `firmware_v1.2.bin` 4. Click **Submit**	Upload bar, “Firm upload versio
TC-ENG-020	Engineer	ISP Config (Edit)	1. Open **Engineer → ISP Config** 2. Modify DNS field 3. Save	Update displa `200 OK`
TC-ENG-030	Engineer	Admin-Only Feature – Parent-Control	1. Directly request `https://…/settings/parent-control` 2. Observe response	Access redire dashb
TC-NEG-001	None (Unauthenticated)	Any Protected Page	1. Open `https://…/settings/firewall` without logging in	Login displa origin prese auth
TC-NEG-002	Engineer	Attempt to modify Admin-only rule via API	1. Capture Admin-only API token (none) 2. Send `PUT /api/firewall` with payload	API re `401 Un`

Pass/Fail Measurement Criteria

Criterion	Pass	Fail
UI Visibility	Element present & enabled	Element missing,

		disabled, or hidden
HTTP Response	Status	Any other status (e.g., `200` for allowed actions, `403/401` for denied actions)
	`200` for allowed actions,	`200` on denied action)
	`403/401` for denied actions	
Backend Verification	DB/Config file reflects change only when role authorized	Change appears despite insufficient role
Audit Log	Log entry records *user, action, outcome* correctly	Missing or incorrect log entry
Session Integrity	Token tied to role; role change requires re-login	Token persists after role downgrade

A test **fails** when **any** of the above criteria are not met for the given TC.

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8. Test Execution